

Jake Walklate

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CAREER SUMMARY

Penultimate dual-degree student seeking an engineering or research-focused role where I can utilise my core cross-disciplinary systems knowledge. Passionate about creating web and embedded software, prototyping physical hardware, and analysing biological data. Previous industry experience writing software code and a proven ability to manage and take ownership of technical decisions and demonstrate a collaborative nature.

I have a rare and technically challenging interdisciplinary skill-set, consisting of medical science and computer related engineering. The added value is in my ability to fluidly translate and collaborate across and between independent and siloed teams needing support, contribute research principles, build end-to-end data processing systems, and providing engineering design and development.

EDUCATION

University of Technology Sydney

Ultimo, NSW

Bachelor of Engineering (Honours) in Artificial Intelligence & Mechatronics

2021 – Jun 2028 (exp.)

- Distinction in Mathematical Modelling
- High Distinction in Computer Programming
- High Distinction in Mechatronic Engineering
- Distinction in Engineering Projects

Bachelor of Medical Science in Pathology

2020 – Jun 2028 (exp.)

- Distinction in Biomedical Polymers

TAFE NSW

St Leonards, NSW

Certificate IV in Cyber Security

Graduated December 2020

Certificate IV in Software Development

Graduated December 2019

EXPERIENCE

Founding Engineer

Mar 2021 – Jan 2022

Coreable

Sydney (Remote)

- Lead the backend and co-developed a full-stack hiring/soft-skills assessment platform used by hundreds of university students in a successful MVP trial.
- Designed and implemented the backend (GraphQL Apollo Server, Node.js, MySQL) with authentication, data models, and non-blocking async architecture.
- Contributed to front-end React.js component development with TypeScript and SCSS using stateful component design principles.
- Collaborated with a small team of transdisciplinary UI/UX designers, stakeholders and entrepreneurs, including academic mentor Sue Dean and facilitator/investor Dirk Vos, on the team which was winners of the 2018 UTS Startups hatchery award.
- Worked in an Agile environment: participated in sprints, code reviews, GitHub workflow ownership, Docker test environments, and GCP-based deployment via CI/CD pipelines.

Programming Analyst

Jun 2019 – Feb 2020

ClockOn

West Gosford, NSW

- Responsible for the technical execution of a major enterprise application rewrite from a legacy Delphi windows application to a modern HTML/C# ASP.NET web stack, by following web accessibility guidelines and modern UX recommendations.
- Developed and implemented engineering solutions using JavaScript and third-party UI components (Vue.JS), with focus on cross-browser support and accessibility, collaborating with project stakeholders in an Agile methodology.

PROJECTS

Custom 3D Printer Upgrade | C++, Python

- Engineered a dual-zone thermal regulation system for an Anycubic Kobra S1 3D printer by utilising a parallel 24V heating array and high-pressure CPAP centrifugal blower to improve vertical polymer adhesion.
- Developed non-blocking and self-configuring C++/Arduino firmware on an ESP32 MCU featuring auto Wi-Fi reconnection handling, mDNS service broadcast and discovery, and a RESTful HTTP endpoint for remote telemetry and automated heating control.

- Repaired a broken camera egress function by reverse engineering compiled C binaries on a Rockchip SoC using Ghidra and reusing shared object libraries in a custom cloud pusher script.
- Ported DOOM to the onboard touch display.

International Space Station Tracker | C++, Arduino

- Modelled a 3D printable robot arm for the purpose of housing an Arduino based IoT project using Autodesk Fusion (360), and selected appropriate polymer materials (such as PA-6/12) and greases (PTFE) for bearings for lowest friction and wear, and ASA for tough and UV resistant structural bodies.
- Called external HTTP APIs to find ISS TLE data, having hard-coded backups in case of network failure, then gathered positional data by using GPS and calculating result with an external SGP4 library, combining orbital mechanics with precision motor control to create a physical pointer that follows the ISS across the sky. When the station passes overhead, the device rotates to track its position in real-time.

Multilayer Perceptron Classifier | Python

- Created an Artificial Neural Network (MLP) from scratch with a bipolar activation function, gradient descent backpropagation and manual momentum optimisation from scratch using only NumPy and no further libraries or plugins for classification of digit inputs.

Blood Cell Classifier | TensorFlow

- Used a Kaggle dataset to build a convolutional neural network, and a multiplayer perceptron classifier with TensorFlow's sequential API and one-hot encoding to automate and distinguish different types of blood cells under microscopy.

Computational Simulation of Human Genetic Inheritance | Rust

- Created computer simulation software to model the heritability of genetic traits involved in partner attraction in humans to determine the percent wise contribution of social and environmental factors to sexual behaviour.
- Used Rust as the language to avoid garbage collectors'/memory overhead to model the earth's population to 1.3 billion individuals on 16GB of RAM, determining an equilibrium of genetic inheritance to contribute to ongoing psycho-sociological research.

OTHER EXPERIENCE

Team Lead (Shift Supervisor)

Pattysmiths Burgers

Sep 2024 – Current
Darlinghurst, NSW

- Contribute to a fast-paced restaurant environment, cooking customer orders, and cleaning.
- Voted most liked team member by management and peers.

Fiber & Ethernet Network Cabler

Mirait Technologies

2015
Sydney, NSW

- Spliced fiber optic cables and connected houses to the National Broadband Network rollout.
- Debugged network faults using testing tools and repaired conduit and terminations.

VOLUNTEER WORK

Member & Contributor

Tech Shed Connections

May 2026 – Current
Canada Bay Library

- Volunteer for the Australian Men's Shed Association, participating in technology conferences, membership meetings, and educational sessions.
- Teach Linux skills, repair vintage tech hardware, contribute to woodworking, and configuring software with vulnerable and retired engineers and technical workers.
- Build 3D printed fighting robots with ESP32 microcontrollers for the purpose of running regular educational sessions for local school children to teach them about STEM projects.

MEMBERSHIPS

Engineers Australia | Student Associate

Google Developer Group Sydney | Member

REFERENCES

Available on request.